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Configurable, power supply voltage referenced single-ended signaling with ESD protection

Abstract

A single-ended data transmission system transmits a signal having a signal voltage that is referenced to a power supply voltage and that swings above and below the power supply voltage. The power supply voltage is coupled to a power supply rail that also serves as a signal return path. The signal voltage is derived from two signal supply voltages generated by a pair of charge pumps that draw substantially same amount of current from a power supply.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to single-ended signaling.

(2) Single-ended signaling is commonly used to transmit electrical signals over wires, such as those in contemporary memory systems. In single-ended signaling, a signal represented by a varying voltage is transmitted over a wire. The voltage varies above or below a reference voltage, which is used by a receiver of the signal to determine the digital values represented by the signal. In conventional single-ended signaling, the reference voltage is typically generated, either on-chip or off-chip, to be at the middle of a signal swing.

(3) Conventional single-ended signaling typically has poor efficiency and limited speed. More specifically, conventional single-ended signaling has the disadvantage that (i) the reference voltage has to be generated for the receiver, (ii) it is difficult to match the reference voltage at the receiver with the reference voltage of the transmitter, (iii) noise such as simultaneous switching noise can be generated due to the use of a power supply voltage for signaling, and (iv) it is difficult to control a signal return path, resulting in crosstalk noise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure herein is illustrated by way of example and without limitation in the figures of the accompanying drawings and in which like reference numerals refer to similar elements and in which:

(2) FIGS. **1**A, **1**B, **1**C, and **1**D illustrate a data transmission system employing power supply voltage referenced single-ended signaling, according to one embodiment.

(3) FIGS. **1**E and **1**F illustrate a cutaway view of an IC package including an integrated circuit of the data transmission system of FIGS. **1**A through **1**D on a printed circuit board.

(4) FIGS. **2**A, **2**B, **2**C, **2**D, **2**E, and **2**F illustrate embodiments of signal voltage supplies employed by the data transmission system of FIGS. **1**A through **1**D.

(5) FIGS. **3**A and **3**B illustrate receiver amplifiers suitable for use in the data transmission system of FIGS. **1**A through **1**D, according to one embodiment.

(6) FIG. **3**C is an alternative embodiment of a common-gate amplifier of FIG. **3**B.

(7) FIG. **4**A illustrates a data transmission system employing ground-referenced single-ended signaling with ESD protection, according to one embodiment.

(8) FIG. **4**B illustrates a voltage swing of a signal in the data transmission system of FIG. **4**A, according to one embodiment.

(9) FIG. **5**A illustrates a configurable data transmission system configured in a single-ended signaling mode, according to one embodiment.

(10) FIG. **5**B illustrates a configurable data transmission system configured in a differential signaling mode, according to one embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

(11) The Figures and the following description relate to embodiments by way of illustration only. Alternative embodiments of the structures and methods disclosed herein may be employed without departing from the principles of the present disclosure.

(12) Embodiments of the present disclosure include a data transmission system using a single-ended signaling system in which a signaling voltage is referenced to a reference voltage that is a power supply voltage shared by a transmitter and a receiver. The signaling voltage swings above and below the reference voltage. The reference voltage may be ground voltage, generally taken to be the lowest power supply potential. The signaling voltage is derived from a first signal supply voltage and a second signal supply voltage, which may be generated by a pair of charge pumps that draw substantially the same amount of current from a power supply, regardless of the output drive level of the transmitter.

(13) FIGS. **1**A, **1**B, **1**C, and **1**D illustrate a data transmission system **100** employing supply voltage referenced single-ended signaling, according to one embodiment. The data transmission system includes first and second integrated circuit devices (ICs) **100***a* and **100***b* coupled to each other via a plurality of conducting lines including signal line **132**. ICs **100***a* and **100***b* may be separate devices with separate IC packages, or combined within a die-stack or other multi-die package, or implemented by bare die. In the case of a multi-die package, the conducting lines may be implemented by intra-package conductors such as wire-bonds, flex conductors or any other suitable interconnection structures. In the case of separately packaged ICs or bare die, the conducting lines may be implemented partly by conductive structures for conveying signals (and establishing supply-rail connection) through the package to an external interconnect, and partly by printed circuit board (PCB) traces, PCB supply bus layers, flex cables or other conductive interconnects between the IC devices. With regard to system function, the ICs may be, for example and without limitation, a processor and chipset (or bridge or other application-specific IC (ASIC)), memory controller and memory device, or any other pair of ICs that carry out operations involving chip-to-

chip signaling.

(14) Still referring to FIG. **1**A, IC device **100***a* is to be connected to a power supply. IC device **100***a* includes a transmitter **102** and terminals (such as pads or pins) **133**, **134**. Transmitter **102** is coupled between two signal supply voltages V**1** and V**2**, which are provided by one or more voltage sources **140**. In one embodiment, V**1** and V**2** are substantially symmetrical with respect to a power supply voltage Vp associated with the power supply (i.e., V**1**−Vp=Vp−V**2**, at least approximately). Terminal **134** is coupled to a power supply rail **130** which is coupled to power supply voltage Vp, while terminal **133** is coupled to signal line **132**. By this arrangement, when input data **106** is applied to transmitter **102**, a signal that swings between about ½V**1** and about ½V**2** according to the input data **106** is output onto signal line **132** via terminal **133**. Thus, transmitter **102** forms a voltage mode transmitter that generates a data signal **120** having a signal voltage that swings substantially symmetrically about the power supply voltage Vp (i.e., between about ½V**1** and about ½V**2**) to convey the bits of the input data **106**.

(15) Because the transmitted signal swings symmetrically about the power supply voltage Vp, the power supply voltage may be used as a signaling reference voltage. In the embodiment shown, for example, the signal line **132** and the power supply rail **134** are coupled, via terminals **136** and **138**, to respective inputs of a receiver **104** in circuit **100***b*. Receiver **104** amplifies the time-varying difference between the signal voltage and the power supply voltage to produce the received data signal (RData) that corresponds to the originally transmitted data **106**. That is, receiver **104** compares the signal received via terminal **136** with the reference voltage supplied via terminal **138** to recover the transmitted signal and generate RData. For example, RData is “1” when the signaling voltage at terminal **136** is higher than the reference voltage at terminal **138**, and RData is “0” when the signaling voltage at terminal **136** is lower than the reference voltage at terminal **138**.

(16) Bi-directional signaling may be effected by including in IC device **100***b* a transmitter **102′** similar to transmitter **102** and in IC device **100***a* a receiver **104′** similar to receiver **104**. The reverse-direction transmitter/receiver **102′**/**104′** may use the same signaling line **132** and/or the same supply-rail signaling reference **130** as transmitter/receiver **102**/**104** (thus driving signaling line **132** bi-directionally). Or, the reverse-direction transmitter/receiver **102′**/**104′** may drive a separate signal line, and/or employ a separate or different supply-rail signaling reference. Also, while not specifically shown within FIG. **1**A, a mesochronous or plesiochronous timing arrangement may be used to enable synchronous data transmission and reception over link **132**. For example, a clock signal may be provided to both IC devices **100***a* and **100***b* (or forwarded from one device to the other) to provide timing for delivery of the input data **106** to the transmitter **102**, and to provide timing for latching of the output of receiver **104** within an input data register or like storage circuit. Alternatively (or additionally), a source synchronous timing reference (e.g., a strobe signal that transitions to mark the presence of data on the signaling line **132**) may be output from the transmitting IC to the receiving IC to control the signal sampling time within the receiving IC. Various timing vernier circuitry may be provided and calibrated to establish data output timing and/or data sample timing within desired margins.

(17) In one embodiment, the V**1**/V**2** signaling supply voltages are selected to effect a small signal swing Vs (e.g., Vs≈½(V**1**−V**2**) is substantially lower than corresponding on-chip logic levels) and thereby enable low-power signaling. The signaling supply voltages may be driven to programmably selected or designed setpoint voltages (as discussed below), or may be adaptively established or calibrated to achieve a threshold operating margin as determined by bit error rate or other signal quality metric.

(18) Charge-pump-based voltage supplies or any other supply voltage circuits or voltage regulators may be used to implement the one or more voltage sources **140**, as discussed in more detail below. In one embodiment, for example, the voltage source(s) **140** is designed such that signal supply voltages are regulated and the net current drawn by transmitter **102** from the power supply remains substantially constant regardless of the output voltage level at terminal **133**, thereby reducing

switching noise within the signaling system.
(19) The signaling arrangement of FIG. **1**A provides a number of benefits relative to conventional approaches. First, the power supply rail **130** may be implemented as a continuous conducting plane (e.g., power plane) that underlies all or most of the signal lines within an IC package or PCB, and thus constitutes a low and sometimes lowest impedance network in the signaling system **100**. Consequently, by coupling the reference output terminal **134** of transmitter **102** and the reference input terminal **138** of receiver **104** to this low impedance reference node (the power plane) **130**, a stable, consistent reference level may be ensured between various components of the signaling system, including transmitter **102** and receiver **104**.
(20) Another advantage of signaling system **100** is that, as discussed below with reference to FIG. **1**C, a signal current sourced by transmitter **102** flows over signaling line **132** to receiver **104** and then returns to transmitter **102** on power plane **130** via a path substantially parallel to the signaling line **132**. Thus, the return current may flow anti-parallel to the signal current, and the signal current and signal-return connections at the chip level are essentially differential. Consequently, noise and cross-talk that would otherwise be generated by return current flow that is not parallel to the signal current flow can be avoided. Altogether, the various benefits of the above-described signaling arrangement enable construction of a high-speed, ultra-low-power single-ended signaling system that requires on the order of 1 mW of signaling energy from the power supply—roughly 10-20 times lower than that of a conventional single-ended signaling system.
(21) FIG. **1**B illustrates an embodiment where a ground plane is used as power supply rail **130**, so that V**1**=+Vs and V**2**=−Vs. Voltage source(s) **140** is shown as comprised of a voltage source **141** that provides +Vs and a voltage source **142** that provides −Vs. Transmitter **102** may include signal-controlled switching elements **108**, **110** each coupled between a common drive node and respective signal supply voltages, +Vs and −Vs. Transmitter **102** further includes a resistive element **114** coupled between the common drive node and terminal **133** to act as a series source termination (e.g., to match a characteristic impedance, ZO, of the signal line). Thus the transmitter's internal ZO impedance **114** and the impedance of the line **132**, terminated at ZO by termination impedance **116**, form a voltage divider that splits the two supply voltages, so that the signaling voltage of signal **120** on line **132** toggles between about +Vs/2 and −Vs/2, as shown in FIG. **1**B. In some embodiments, the functions of switches **108**, **110** and resistor **114** may be combined in a single device, for example an MOS field effect transistor (MOSFET) operating in the linear or “triode” region of operation; in such embodiments, the linear relationship of the voltage between the drain and source terminals of the MOSFET and the current flowing between drain and source causes the device to operate in a mode approximating a resistor.
(22) Still referring to FIG. **1**B, receiver **104** may include a differential amplifier or other comparator circuit **112**. Amplifier **112** amplifies the time-varying difference between the ground rail and signal potentials to produce the received data signal (RData) that corresponds to the originally transmitted data **106**. As shown, a termination element (depicted as resistive element **116**) may be coupled between the input nodes **136** and **138** to terminate the incoming signal line according to the characteristic impedance, ZO.
(23) Referring now to FIG. **1**C, a signal current **150** sourced by transmitter **102** flows over signaling line **132** to receiver **104** and then returns to transmitter **102** on ground plane **130** via a path that can be made substantially parallel to the signaling line **132**. Thus, noise and cross-talk that would otherwise be generated by return current flow not parallel to the signal current flow can be reduced or eliminated.
(24) In one embodiment, IC devices **100***a* and **100***b* include IC chips within respective semiconductor packages **184** each having an array of solder balls or other metal contacts that act as terminals for the IC devices. FIG. **1**D illustrates IC devices **100***a* and **100***b* on a PCB **101**, according to one embodiment. FIG. **1**D shows two signaling lines **180**, **182** between circuits **100***a*, **100***b*, as a subset of a plurality of signaling lines that would be present on the PCB **101** between

circuits **100***a* and **100***b*. IC device **100***a* includes a plurality of transmitters, including transmitters **102***a*, **102***b*. IC device **100***b* includes a plurality of receivers, including receivers **104***a*, **104***b*. IC devices **100***a* and **100***b* include a plurality of reference (GND) terminals **164**, **168** and **174**, **178**, respectively, that are coupled to a power plane (e.g. ground plane) **131** of the PCB **101**. The power plane **131** of the PCB **101** can be a solid metal layer in the PCB **101**, a metal layer with windows or holes, or it can include a plurality of conducting lines each coupled to a same power supply voltage. Although FIG. **1**D shows that power plane **131** is between IC devices **100***a* and **100***b*, in practice, power plane **131** may extend under IC devices **100***a* and **100***b*.

(25) IC devices **100***a* and **100***b* further include a plurality of signaling terminals **162**, **166** and **170**, **174**, respectively. To facilitate routing of the signaling lines, the solder balls acting as or coupled to the signaling terminals **162**, **166** and **170**, **174** can be placed closer to the edge of the respective semiconductor packages **184** than the solder balls acting as or coupled to the reference (e.g., GND) terminals **164**, **172** and **168**, **178**. For example, in the IC packaging **184** of circuit **100***b* (as shown by the insert in FIG. **1**D), the signaling terminals **170**, **174** are placed as outer balls of the IC packaging **184** to facilitate easy connection of the balls to the signaling lines **180** and **182** outside the IC package. On the other hand, the reference terminals **172**, **174** can be connected to the inner balls of the IC packaging **184**, because they only need to be coupled to a power plane (e.g., GND plane) within the package, which is coupled to the power plane **131** of the PCB **101**, to which the package is soldered or otherwise attached. Such ground package pins **172**, **178** are not as expensive as signal pins **170**, **174**, since they are connected directly through vias to a solid conducting plane, thereby requiring no trace routing. Also, although FIG. **1**D illustrates that separate reference (e.g., GND) terminals **164**, **172** and **168**, **178** are used for each of the corresponding signaling terminals **162**, **170** and **166**, **174**, it is also possible for two or more of the signal pins **162**, **166** and **170**, **174** to share a common reference terminal, thereby further reducing the number of package pins needed for the reference pins.

(26) FIG. **1**E illustrates a cutaway view of an IC package including IC device **100***a* (or IC device **100***b*) on a printed circuit board. According to one embodiment, referring to FIG. **1**E, IC device **100***a* is mounted on a laminated IC package substrate **200**. Package substrate **200** is a multi-layer package substrate having signal wiring conductive layers, including a top layer (e.g., signal trace **202**) and a bottom layer (e.g., solder-ball pads **206***a* and **206***b*). Package substrate may also include additional conducting layers separated from each other and from the top and bottom layers by insulating layers **204**. The additional layers may used for power delivery and signal return current conduction. For example, layer **201** in FIG. **1**E may be a signal return (and/or reference) plane for the previously described single-ended signaling system, while layer **205** may be a plane used to connect to some other power supply voltage. In one embodiment, reference plane **201** corresponds to the ground reference plane **130** of FIG. **1**B (i.e., coupled to an external ground reference) and is sometimes referred to as ground plane **201** below. In alternative embodiments, reference plane **201** may be coupled to another DC reference or even a time-varying reference.

(27) Still referring to FIG. **1**E, integrated circuit package **200** may be attached both electrically and mechanically to printed wiring board **190** via an array of solder balls, including solder balls **207***a* and **207***b*. Printed wiring board **200** may be constructed similarly or identically to laminated package substrate **200**, with surface conducting layers, such as layer **191**, inner conducting planes, such as plane **194**, and perhaps additional internal signal layers and power supply planes, that are separated from each other by insulating layers **193**.

(28) Still referring to FIG. **1**E, IC device **100***a* includes one or more instances of ground-referenced signaling receivers and transmitters, such as **102** and **104′** in FIG. **1**A. The output pads of these units (corresponding to **133** and **134** in FIG. **1**A) are connected to conducting traces and planes on package **200** via wire bonds **188** and **189**, respectively. In FIG. **1**E, the signal terminal **133** is connected to package trace **202** via wire-bond **188**. Signal current, denoted by the dashed lines, flows from the chip **100***a*, through wire-bond **188** to package trace **202**, thence to through-package

via **203**, where it is carried to the bottom layer of the package to solder-ball pad **206***a*. Signal current further flows through solder ball **207***a* to printed wiring board solder-ball pad and signal trace **191**, and from there to another packaged integrated circuit such as IC device **100***b*. Return current (also shown by a dashed line) flows back from the second packaged integrated circuit over a power supply plane **194**, to via **195**, where it is conducted up to solder-ball pad **192**, through solder ball **207***b*, through solder-ball pad **206***b* and via **208** to the on-package conducting ground plane **201**. The electromagnetic interaction between the signal current and this return current will tend to cause the return current to flow in plane **201** in a manner anti-parallel to the signal current in signal trace **202** (i.e., reverse-direction current flow in a path parallel to the signal trace **202**), finally reaching integrated circuit **100***a* through wire bond **189**. In one embodiment, the solder balls **207***a*/**207***b*, solder-ball pads **206***a*/**206***b*, and through-package vias **203**, **208** that carry the signal current and return current are disposed in close proximity. The close proximity of the solder balls, solder-ball pads, and through-package vias that carry the signal current and return current help avoid large series inductances from accumulating in the overall signal path, and thus further reduces or eliminates cross-talk with neighboring signals and their return currents.
(29) FIG. **1**E is shown merely by way of example, as there are many other ways of implementing an integrated circuit package and printed wiring boards, including, but not limited to "flip-chip" (C4) construction, package-on-package (POP), and other packaging constructions. Although the technology used and the configuration of the package may vary, the following features should preferably be maintained for optimal performance: (1) the use of a common, low-impedance power supply conducting plane (e.g., power supply conducting plane) that carries all return currents from one or more supply-reference single-ended signaling components (i.e., transmitters and receivers); and (2) provision for a separate path from the conducting plane to the reference terminals of both transmitters and receivers (e.g., pins **134** and **138** in FIG. **1**A), which path may include, for example and without limitation, board and package conducting planes, solder balls, through-package and through-board vias, wire bonds, and C4 solder bumps. Furthermore, to the extend the extent permitted by the packaging construction technique, the signaling system may also include an arrangement that forces the signal return current to flow anti-parallel to the corresponding signal current, while not overlapping with the signal current return path of other signaling components.
(30) FIG. **1**F illustrates yet another packaging embodiment, similar to the embodiment of FIG. **1**D (and including all the variants mentioned above), but in which the ground plane **201** extends beneath the IC die **100***a*. In the embodiment shown, the IC die **100***a* is insulated from the ground plane by a relatively thin layer of epoxy **197**. Other insulating materials may be used in alternative embodiments. This approach generally provides all the benefits of the arrangement of FIG. **1**E while maintaining the presence of the ground plane **201** beneath the IC die and thus providing potentially superior shielding and noise suppression.
(31) FIG. **2**A illustrates one embodiment of the voltage source(s) **140**. As shown in FIG. **2**A, voltage source(s) **140** includes charge pumps **212** and **214** for generating the signal supply voltages +Vs and −Vs, respectively, and a charge pump regulator **210**. Regulator **210** is coupled to the signal supply voltages +Vs and −Vs, and receives a reference voltage Vref that represents a desired voltage level for Vs. It controls or regulates the charge pumps **212** and **214** using control signals ϕ**2***b* and ϕ**2***a*, respectively. Charge pumps **212** and **214** are each coupled between first and second power supply voltages (e.g., Vdd and GND), and are configured to draw current from the power supply to provide the signal supply voltages +Vs and −Vs, and to maintain the signal supply voltages +Vs and −Vs at desired levels in response to control signal ϕ**1** and to control signals ϕ**2***b* and ϕ**2***a*, respectively.
(32) FIG. **2**B illustrates an embodiment of regulator **210**. As shown in FIG. **2**B, regulator **210** receives a pair of non-overlapping clocks ϕ**1** and ϕ**2** (or, it may include a clock signal source **220** that generates the pair of non-overlapping clocks ϕ**1** and ϕ**2**). A pair of equally sized resistors **216** and **218** divides a voltage difference between +Vs and −Vs to produce a voltage Vcm (a common-

mode voltage between the positive and negative signal supply voltages). The signal supply voltages +Vs and −Vs are nominally symmetric about GND. A digital or "bang-bang" control loop including sense amplifier/latch **224** amplifies the difference (error) voltage between Vcm and GND. When the voltage on Vcm is higher than the voltage on GND, a signal on line **228** is asserted, and during the following ϕ**2** interval, AND gate **234** outputs signal ϕ**2***a* that tracks signal ϕ**2** (i.e., signal ϕ**2***a* is asserted if signal ϕ**2** is asserted). On the other hand, if the voltage on Vcm is lower than the voltage on GND, sense amplifier/latch **224** amplifies this difference and de-asserts the signal on line **228**, such that ϕ**2***a* is not asserted.
(33) Likewise, digital or "bang-bang" control loop including sense amplifier/latch **222** that amplifies the difference (error) voltage between +Vs and Vref. If the voltage on +Vs is lower than the voltage Vref, a signal on line **226** is asserted, and during the following ϕ**2** interval, AND gate **232** outputs signal ϕ**2***b* that tracks signal ϕ**2** (i.e., signal ϕ**2***b* is asserted if signal ϕ**2** is asserted). On the other hand, if the voltage +Vs is higher than the voltage Vref, sense amplifier/latch **222** amplifies this difference and de-asserts the signal on line **226**, such that ϕ**2***b* is not asserted. By asserting and deasserting ϕ**2***a* and ϕ**2***b* based on comparison of Vcm with GND and +Vs with Vref, respectively, the signal supply voltages +Vs and −Vs can be maintained at the desired levels, as discussed below.
(34) FIG. **2**C illustrates another embodiment of regulator **210**. Here, the signal supply voltage +Vs is fed back into comparator **242** for comparison with the reference voltage Vref. The output **246** of comparator **242** is asserted if +Vs is smaller than Vref. Thus, the output ϕ**2***b* of AND gate **252** follows the clock signal ϕ**2** when +Vs is lower than Vref and is not asserted otherwise. Likewise, the signal supply voltage −Vs is fed back into window comparator **244** for comparison with the reference voltage Vref. The output **248** of window comparator **244** is asserted if the difference between GND and −Vs is smaller than the difference between Vref and ground (i.e., GND−(−Vs)<Vref−GND, or Vs is smaller than Vref). Thus, the output ϕ**2***a* of AND gate **254** follows the clock signal ϕ**2** when the difference between GND and −Vs is smaller than the difference between Vref and GND. Otherwise, it is not asserted.
(35) FIGS. **2**D and **2**E illustrate embodiments of charge pumps **212** and **214**, respectively. Referring to FIG. **2**D, charge pump **212** includes a switching device **262** (shown as a P-type metal-on-semiconductor (MOS) field effect transistor (PFET)) that is turned on or off according to clock signal ϕ**1**, and a switching device **264** (shown as an N-type MOS field effect transistor (NFET)) that is turned on or off according to clock signal ϕ**2***b*. Charge pump **212** further includes a capacitor **266** and a capacitor **268**. Capacitor **266** has a much smaller capacitance than that of capacitor **268** and is therefore sometimes referred to as Csmall, while capacitor **268** is sometimes referred to as Cbig. Cbig **268** is coupled between +Vs and GND, so the voltage across capacitor **268** is the signal supply voltage +Vs.
(36) FIG. **2**F is timing chart showing signals ϕ**1**, ϕ**2**, ϕ**2***b*, the voltage across capacitor **266** (V_Csmall), and the voltage across capacitor **268** (+Vs) in comparison with Vref during a series of time periods after IC device including the charge pumps is turned on. As shown in FIG. **2**F, during time period t**1**, when ϕ**1** is asserted, switch **262** is turned on, allowing charge to flow from Vdd to capacitor **266**, so that a voltage across capacitor **266**, V_Csmall, increases to a level between Vdd and GND. The voltage across Cbig is lower than Vref so ϕ**2***b* is tracking ϕ**2**. During time period t**2**, when ϕ**1** is de-asserted while ϕ**2** and thus ϕ**2***b* is asserted, switch **262** is turned off while switch **264** is turned on, so that stored charge in Csmall is shared with Cbig and the voltage across capacitor **268**, +Vs, increases while V_Csmall decreases.
(37) Since the capacitance of Csmall is much smaller than the capacitance of Cbig, and the increase in voltage across Cbig during time period t**2** is much smaller than the decrease in voltage across Csmall, so the voltage across Cbig at the end of time period t**2** is slightly higher than the voltage across Cbig at the start of time period t**2**. The above charging of Csmall and subsequent charging of Cbig may repeat in response to signals ϕ**1** and ϕ**2***b*, until the signal supply voltage +Vs reaches or is

higher than Vref at the end of time period t**3**. As discussed above, as long as +Vs remains at or above Vref during time periods t**4** and t**4**, ϕ**2***b* is not asserted, meaning switch **264** does not turn on and Cbig does not get further charged by Csmall. When +Vs drops below Vref during time period t**6**, ϕ**2***b* follows ϕ**2** again in time period t**7**, and charges from Csmall will be shared with Cbig until +Vs is higher than Vref again.

(38) The signal and voltage curves shown in FIG. **2**F are for illustrative purposes only and are not to replicate or scale with signals or voltages in a real IC device. In some embodiments, the voltage between Vdd and GND is much higher than Vref. For example, Vdd may be about 1 volt, while Vref is about 0.1 volt. In such case, if the charge pumps run at a frequency of, for example, 1 GHz, data is being driven out at, for example, 4 Gbps, and the impedance of transmitter **102** (thus the impedance of the load) is, for example, 50 ohms, Csmall can be about 0.75 pF to about 1.25 pF while Cbig can be about 50 pF to about 250 pF.

(39) On the other hand, charge pump **214** includes switches **272**, **282** that are turned on or off according to clock signal ϕ**1**, and switches **274**, **284** that are turned on or off according to clock signal ϕ**2***a*. Charge pump **214** further includes a capacitor Csmall **276** and a capacitor Cbig **278**. Cbig **278** is coupled between GND and −Vs so the voltage across Cbig **278** is the signal supply voltage −Vs. Switches **272**, **282** are turned on when clock signal ϕ**1** is asserted so that charge is extracted from power supply Vdd via switch **272** and **282** and stored in capacitor Csmall **212**. Switches **274** and **284** are turned on when signal ϕ**2***a* is asserted and signal ϕ**1** is de-asserted, so charges stored in Csmall is pumped into Cbig. This process repeats until the common mode voltage Vcm is lower than GND, or the difference between GND and Vs is larger than the difference between Vref and GND. As discussed above in association with FIGS. **2**B and **2**C, as long as the common mode voltage Vcm remains lower than GND, or the difference between GND and Vs remains larger than the difference between Vref and GND, ϕ**2***a* remains de-asserted and no charge is pumped from Csmall **276** to Cbig **278** to further increase the voltage across Cbig **278**. When the common mode voltage Vcm becomes higher than GND, or the difference between GND and Vs becomes smaller than Vref, ϕ**2***a* follows ϕ**2** again and Cbig **278** gets charged from Csmall until Vcm becomes lower than GND, or the difference between GND and Vs becomes larger than the difference between Vref and GND again.

(40) FIGS. **2**D and **2**E shows that switches **262** and **272** are implemented using P-type metal-oxide-semiconductor field effect transistors (P-MOSFET or PFET), while switches **264**, **274**, **282** and **284** are implemented using N-type MOSFETs (NFET). PFET **262** or **272** is turned on when ϕ**1** is asserted by virtue of inverter **260** or **270**, respectively, which inverts the sense of ϕ**1** to drive the gate terminal of PFET **262** or **272**. Those skilled in the art will easily understand that other types of switches can be used in place of the MOSFETs. The capacitors **266**, **268**, **276**, and **278** may be implemented using MOS capacitors, but other types of capacitors can alternatively be employed.

(41) The embodiments of FIGS. **2**A through **2**F are shown by way of examples. There are many other possible embodiments of on-chip voltage sources or regulators. For example, some embodiments may have additional Csmall capacitors and additional switches that charge these Csmall capacitors in a series circuit arrangement and discharge the Csmall capacitors in a parallel arrangement, thereby providing higher overall efficiency than that of the simple charge pumps of FIGS. **2**D and **2**E. In a further example, instead of using clocks to gate the MOSFETs on and off in the charge pumps, other type of switches that respond to other characteristics of control signals such as frequency, amplitude or duty cycle of the control signals may be used. In yet another example, switched capacitors may be used in place of the switch and capacitor combinations in the charge pumps. In still other examples, other regulating means such as a linear regulator or switching regulator may be used to control the signal supply voltages.

(42) In one embodiment, charge pumps **212** and **214** are implemented with components having substantially the same efficiency. In the embodiment shown in FIG. **2**A, charge pump **212** draws current from Vdd when transmitter **102** is transmitting “1”s while charge pump **214** draws current

from Vdd when transmitter **102** is transmitting “0”s. Thus, assuming that the transmitter **102** has matching impedance with the transmission lines **132**, **130**, the transmitter **102** draws approximately the same amount of supply current from the power supply regardless of which way the transmitter **102** is driving the current (i.e., sourcing current from transmitter **102** to receiver **104**, or sinking current from receiver **104** to transmitter **102**). Thus, to the extent that charge pumps **212** and **214** have about the same efficiency, the complementary configuration of the voltage source(s) **140** ensures that supply current variation in signaling system **100** is near zero, thereby reducing or eliminating simultaneous switching noise (SSO) inherent in conventional single-ended signaling schemes.

(43) FIGS. **3**A, **3**B, and **3**C illustrate several alternative embodiments for the receiver amplifiers used in the data transmission system **100** of FIGS. **1**A, through **1**D, when the GND is chosen as the reference voltage, according to one embodiment. FIG. **3**A illustrates a PMOS amplifier **302** that can be used in some embodiments of receiver **104**, coupled to the two terminals **136**, **138** corresponding to the signaling voltage and GND reference voltage, respectively. This amplifier is a conventional PMOS differential amplifier with resistor loads. Numerous alternative embodiments of a PMOS differential amplifier are possible. FIG. **3**B illustrates one embodiment of a common gate amplifier **304** that can be used in receiver **104** in other embodiments, coupled to the two terminals **136**, **138** corresponding to the signaling voltage and GND reference voltage, respectively. The common gate amplifier **404** in FIG. **3**B is implemented with NMOS transistors and thus may have a higher gain-bandwidth product than the PMOS amplifier **402** of FIG. **3**A.

(44) FIG. **3**C is a common-gate amplifier **306** according to another alternative embodiment. Current source **906** establishes a reference current in NFET **914**, thereby generating gate reference voltage Vcas. NFETs **910** and **912**, which are the amplifying transconductances in amplifier **306**, are drawn at the same shape factor (width/length) as **914** so that they carry about the same current as **914**, when their inputs, driven into their source terminals, attached to pins **136** and **138**, are at substantially the same potential, since in this case the gate-to-source voltages (Vgs) for all three transistors are substantially equal and the shape factors are identical. The data signal, attached to pin **136**, is compared in amplifier **306** with the signal reference voltage, input on pin **138** from the power supply GND plane **130**. The current from line to GND develops a voltage across termination resistor **920**; this voltage swings from about +Vs/2 to −Vs/2. When the voltage on line (pin **136**) is higher than the voltage on GND (pin **138**) NFET **910**'s Vgs is less than Vgs at NFET **912**. Therefore, less current flows through **910** and through **912**, and the voltage drop through load resistor **902** is smaller than the voltage drop through load resistor **904**. This causes the voltage on outH to rise above the voltage on outL. On the other hand, if the voltage on **136** is lower than the voltage on **138**, then NFET **910**'s Vgs is greater than Vgs for NFET **912**. Consequently NFET **910** carries more current than **912**, and the voltage across **902** is larger than the voltage across **904**. In this case, the voltage on outH is less than the voltage on outL. Under these circumstances the current driven back into line **132** (pin **136**) by NFET **910** is larger than the current driven into the line **132** by **910** when the voltage on line exceeds the voltage on GND. To prevent this current unbalance from introducing an unwanted voltage offset in amplifier **306**, a compensating resistor **922**, RC, is used to return some of the current to the negative power supply −Vs.

(45) Still referring to FIG. **3**C, the magnitude of the voltage between outH and outL is higher than the magnitude of the voltage between line (pin **136**) and GND (pin **138**) thanks to the gain of amplifier **306**; the gain of **306** is about Gm×RL, where Gm is the transconductance of NFETs **910** and **912**, and RL is the resistance of load resistances **902** and **904**.

(46) Still referring to FIG. **3**C, it will be clear to those skilled in the art, that the terminating impedance presented across pins **136** and **138** is the parallel combination of the terminating resistor RT **920**, the compensating resistor RC **922**, and the input impedance of the amplifier, which is about 1/Gm, where Gm is the transconductance of transistors **910** and **912**. This parallel combination of resistances should be adjusted to about the characteristic impedance of the

transmission line formed by line **132** and GND to avoid reflections in the transmission line.
(47) The foregoing embodiments of input amplifiers are shown by way of examples. Many alternative embodiments of input amplifiers can be employed in receiver **104** or **104**′ in system **100**.
(48) FIG. **4**A illustrate a data transmission system employing GND referenced single-ended signaling with ESD protection, according to one embodiment. The data transmission system **400** includes circuit **100***a* including the transmitter **102** and circuit **100***b* including the receiver **104**, and employs the single-ended signaling scheme as described above using the signaling line **132** and GND reference line **130**. Voltage sources such as charge pumps (CP) **141**, **142** supply the positive and negative signal supply voltages V.sub.S1 and −V.sub.S1, and either switch **108** or switch **110** is turned on to source or sink line current on the signaling line **132** responsive to input data **106**, as explained above. The signaling voltage is developed across resistor **116** and detected by input amplifier **112** to recover the input data **106**.
(49) The data transmission system **400** additionally includes ESD (Electrostatic Discharge) protection devices **402**, **404**. In one embodiment, each ESD protection device **402**, **404** is comprised of a pair of cross-coupled diodes connected between the signaling voltage **132** and the GND reference voltage **130** of the single-ended signaling system, with ESD protection device **402** connected on the transmitter side **102** and the ESD protection device **404** connected on the receiver side **104**. Note that the cross-coupled diodes in ESD protection device **404** are coupled between the inputs of the input amplifier **112** in parallel with resistor **116**. The ESD protection devices **402**, **404** have the benefit of reducing the input capacitance of the I/O pins **133**, **134** and **136**, **138** and protecting the pins **133**, **134** and **136**, **138** from electrostatic discharge. Also, as shown in FIG. **4**B, the allowed voltage swing **408** (V.sub.L=−50 mV to V.sub.H=+50 mV) of the signaling voltage **132** should be within the margins **416**, **418** between the regions **406**, **408** in which diodes **402**, **404** become forward biased. Symmetrically connecting the diodes **402**, **404** to the GND signaling reference voltage **130** allows the voltage swing of the signaling voltage on line **132** to be larger than for asymmetric connection, because the margins **416**, **418** between the regions **406**, **410** also become symmetrical with respect to the GND reference voltage **130** and thus no margins **416**, **418** between the two regions **406**, **410** are wasted.
(50) FIG. **5**A illustrates a configurable data transmission system configured in single-ended signaling mode, according to one embodiment. The data transmission system **500** includes circuits **500***a*, **500***b*. Circuit **500***a* includes a transmitter **502** and circuit **500***b* includes a receiver **504**. Transmitter **502** and receiver **504** and employs the ground-referenced single-ended signaling as described above using the signaling line **132** and GND reference line **130**. Charge pumps (CP) **140**, **142** supply the positive and negative signal supply voltages V.sub.S1 and −V.sub.S1 for generating the signaling voltage on line **132**, and either switch **108** or switch **110** is turned on to source or sink line current on the signaling line **132** responsive to input data **106** as explained above. The signaling voltage is detected across resistor **116** and detected by input amplifier **112** to recover the input data **106**. ESD protection devices **402**, **404** each comprised of a pair of cross-coupled diodes are also connected between the terminals **133**, **134** and between the terminals **136**, **138**, respectively.
(51) Circuit **500***a* additionally includes terminal **532** and ESD protection device **522**, and transmitter **502** additionally includes switches **508**, **510** and resistor **544**. Circuit **500***b* additionally includes ESD protection device **524** and terminal **536**, and the receiver **504** additionally includes switches **512**, **514**, **516**, **518**. An additional transmission line **552** is coupled between terminal **532** and terminal **536**. Terminals **133**, **134**, **532** form one configurable unit (set) of pins of the transmitter **502** among a number of output pins that would be present on the IC including the transmitter **502**. Similarly, terminals **136**, **138**, **536** form one configurable unit (set) of pins of the receiver **504** among a number of output pins that would be present on the IC including the receiver **504**. However, when the data transmission system **500** is used in single-ended signaling mode, terminals **532**, **536**, line **552**, ESD protection devices **522**, **524** are not used, because switch **518** is

turned off, disconnecting the input amplifier **112** from terminal **536** and line **552**. Also, switch **512** and switch **518** are turned on, connecting terminals **136** and **138** to the inputs of amplifier **112**. Switch **514** is a dummy transistor for providing matched loading corresponding to switch **518**, but is also turned off. On the transmitter side **502**, switches **508**, **510** and resistor **544** are not used because they are connected to line **552** that is disconnected from input amplifier **512**. Thus, the data transmission system **500** is configured for single-ended signaling mode, similar to the transmission system **400** of FIG. **4**, using signaling line **132** and the GND reference line **130**.

(52) FIG. **5**B illustrates a configurable data transmission system configured in differential signaling mode, according to one embodiment. The components and structure of the data transmission system **500** shown in FIG. **5**B are the same as those of the data transmission system **500** shown in FIG. **5**A. However, the data transmission system **500** in FIG. **5**B is configured to operate in differential signaling mode. Differential signaling transmits data by means of two complementary signals sent on two separate wires. Here, lines **132** and **552** are used as the complementary (positive and negative) signaling lines for differential signaling, and the GND reference line **130** is not used. On the receiver side **504**, switches **512**, **518** are turned on to connect terminals **136**, **536** (and signaling lines **132**, **552**) to input amplifier **112**, respectively, but switch **516** is turned off to disconnect terminal **138** (and the GND reference line **130**) from the input amplifier **112**. Switch **514** is also turned off. On the transmitter side **502**, both pairs of switches **108**, **110** and **508**, **510** are used to generate the complementary differential signals to be sent over transmission lines **132**, **552**. Switches **108**, **110** are connected to signaling line **132** via resistor **114** to generate the positive differential signal (Out+) on line **132**, and switches **508**, **510** are connected to signaling line **552** via resistor **544** to generate the negative differential signal (Out−). Further, both ESD protection devices **402**, **522** are used on the transmitter side **502** and both ESD protection devices **404**, **524** are used on the receiver side **504**. Thus, the data transmission system **500** is configured for differential signaling mode, using both signaling lines **132**, **552** and terminals **133**, **532** on transmitter side **502** and terminals **136**, **536** on the receiver side **504**.

(53) The data transmission system described in FIGS. **5**A and **5**B is configurable to a single-ended signaling system or a differential signaling system in a convenient manner, simply by turning on or off switches **516**, **518**. For example, by adding a pin (e.g., pin **536**) for every pair of pins (pins **136**, **138**) on the receiver side **504**, a signaling interface that is configurable and versatile with high signal integrity can be achieved.

(54) Alternatively, no additional pins are required. In single-ended signaling mode each pair of pins are line pin and reference or GND pin, respectively. In differential signaling mode they are lineP pin and lineN pin, respectively. The lineN half of the transmitter is turned off in single-ended signaling mode, and a large NFET switch can connect what was lineN to internal Vss. A dummy switch on lineP can be added to avoid imbalances. Both terminals would have cross-coupled diodes returned to the GND rail. At the receiver, no switches are required. The lineN input can just be connected to GND. Thus, a configurable signal transmission system similar to that described in FIGS. **5**A and **5**B but without the extra pins **536** and **532** can be formed.

(55) It should be noted that the various integrated circuits, dice and packages disclosed herein may be described using computer aided design tools and expressed (or represented), as data and/or instructions embodied in various computer-readable media, in terms of their behavioral, register transfer, logic component, transistor layout geometries, and/or other characteristics.

(56) When received within a computer system via one or more computer-readable media, such data and/or instruction-based expressions of the above described circuits may be processed by a processing entity (e.g., one or more processors) within the computer system in conjunction with execution of one or more other computer programs including, without limitation, net-list generation programs, place and route programs and the like, to generate a representation or image of a physical manifestation of such circuits. Such representation or image may thereafter be used in device fabrication, for example, by enabling generation of one or more masks that are used to form

various components of the circuits in a device fabrication process.
(57) In the foregoing description and in the accompanying drawings, specific terminology and drawing symbols have been set forth to provide a thorough understanding of the present invention. In some instances, the terminology and symbols may imply specific details that are not required to practice the invention. For example, any of the specific numbers of bits, signal path widths, signaling or operating frequencies, component circuits or devices and the like may be different from those described above in alternative embodiments. In other instances, well-known circuits and devices are shown in block diagram form to avoid obscuring the present invention unnecessarily. Additionally, the interconnection between circuit elements or blocks may be shown as buses or as single signal lines. Each of the buses may alternatively be a single signal line, and each of the single signal lines may alternatively be buses. Signals and signaling paths shown or described as being single-ended may also be differential, and vice-versa. A signal driving circuit is said to "output" a signal to a signal receiving circuit when the signal driving circuit asserts (or deasserts, if explicitly stated or indicated by context) the signal on a signal line coupled between the signal driving and signal receiving circuits. The term "coupled" is used herein to express a direct connection as well as a connection through one or more intervening circuits or structures. Integrated circuit device "programming" may include, for example and without limitation, loading a control value into a register or other storage circuit within the device in response to a host instruction and thus controlling an operational aspect of the device, establishing a device configuration or controlling an operational aspect of the device through a one-time programming operation (e.g., blowing fuses within a configuration circuit during device production), and/or connecting one or more selected pins or other contact structures of the device to reference voltage lines (also referred to as strapping) to establish a particular device configuration or operation aspect of the device. The terms "exemplary" and "embodiment" are used to express an example, not a preference or requirement.
(58) While the invention has been described with reference to specific embodiments thereof, various modifications and changes may be made thereto without departing from the broader spirit and scope. For example, features or aspects of any of the embodiments may be applied, at least where practicable, in combination with any other of the embodiments or in place of counterpart features or aspects thereof. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

Claims

1. An integrated circuit, comprising: a first power supply terminal to receive a first power supply voltage; a second power supply terminal to receive a second power supply voltage; a first terminal; a second terminal to be at the second power supply voltage; a transmitter to transmit a transmit signal via the first terminal, the transmit signal representing a first data stream using a first voltage level and a second voltage level, the first voltage level and the second voltage level to be substantially symmetric above and below the second power supply voltage; and a receiver to receive a second data stream that uses the first voltage level and the second voltage level, the receiver configurable to operate in a single-ended signaling mode where a single-ended signal is received via the first terminal, and a differential-signaling mode in which a differential signal is received via the first terminal and a third terminal.
2. The integrated circuit of claim 1, further comprising: a first electrostatic discharge protection circuit couple between the first terminal and the second terminal and having two non-overlapping forward bias regions.
3. The integrated circuit of claim 2, wherein the first voltage level and the second voltage level are between the two non-overlapping forward bias regions.
4. The integrated circuit of claim 1, wherein the second terminal is part of a signal return path for

the transmit signal.
5. The integrated circuit of claim 1, wherein the second power supply voltage is a ground voltage.
6. The integrated circuit of claim 1, wherein the second power supply voltage is other than a ground voltage.
7. The integrated circuit of claim 1, wherein the receiver comprises a differential amplifier to receive the second data stream.
8. An integrated circuit, comprising: a first terminal; a second terminal to be at a first power supply voltage; a transmitter circuit to transmit a first signal while a first charge pump and a second charge pump draw a total power supply current from a second power supply voltage that is substantially independent of a voltage level of the first signal, the first signal switching between a first voltage level and a second voltage level, the first voltage level and the second voltage level being substantially symmetrical with respect to the first power supply voltage; and a receiver circuit to receive a second signal that switches between the first voltage level and the second voltage level, the receiver circuit configurable to operate in a single-ended signaling mode where a single-ended signal is received via the first terminal, and a differential-signaling mode in which a differential signal is received via the first terminal and a third terminal.
9. The integrated circuit of claim 8, wherein the receiver circuit comprises a differential amplifier to receive the second signal.
10. The integrated circuit of claim 8, further comprising: a first charge pump to draw a first current from a second power supply voltage to generate the first voltage level; and a second charge pump to draw a second current from the second power supply voltage to generate the second voltage level.
11. The integrated circuit of claim 10, wherein the first power supply voltage is a ground voltage.
12. The integrated circuit of claim 10, wherein the first power supply voltage is other than a ground voltage.
13. The integrated circuit of claim 8, further comprising: a first electrostatic discharge protection circuit couple between the first terminal and the second terminal and having two non-overlapping forward bias regions.
14. The integrated circuit of claim 13, wherein the first voltage level and the second voltage level are between the two non-overlapping forward bias regions.
15. A method, comprising: receiving, at a first power supply terminal, a first power supply voltage; receiving, at a second power supply terminal, a second power supply voltage; transmitting, via a first terminal, a transmit signal representing a first data stream using a first voltage level and a second voltage level, the first voltage level and the second voltage level being substantially symmetric above and below the second power supply voltage; and configuring a receiver in one of a first mode and a second mode to receive a second data stream that uses the first voltage level and the second voltage level, the first mode to receive the second data stream via the first terminal, the second mode to receive the second data stream via the first terminal and a third terminal.
16. The method of claim 15, wherein the first mode configures the receiver to operate in a single-ended signaling mode where a single-ended signal is received via the first terminal.
17. The method of claim 16, wherein the receiver comprises a differential amplifier and, in the first mode, the first terminal is coupled to the differential amplifier to receive the second data stream.
18. The method of claim 15, wherein the second mode configures the receiver to operate in a differential-signaling mode in which a differential signal is received via the first terminal and the third terminal.
19. The method of claim 18, wherein the receiver comprises a differential amplifier and, in the second mode, the third terminal is coupled to the differential amplifier to receive the second data stream.
20. The method of claim 15, wherein the first voltage level and the second voltage level are in a non-conducting voltage region of an electrostatic discharge protection circuit.